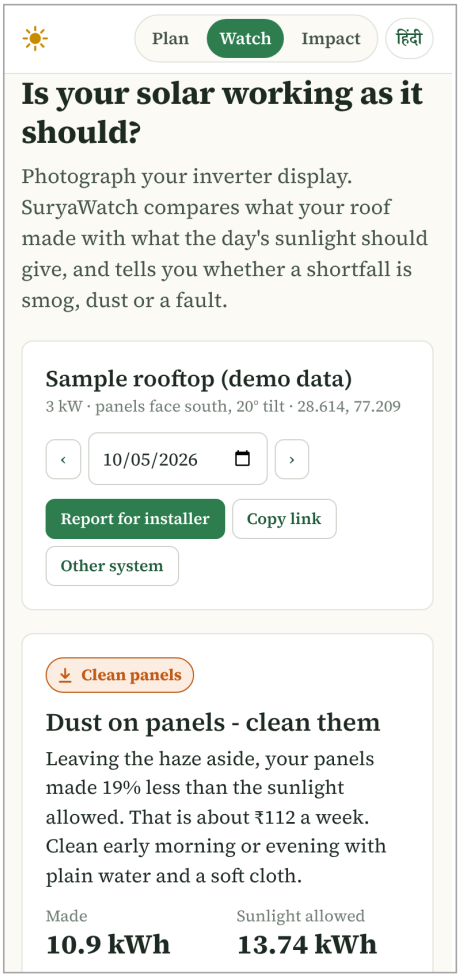
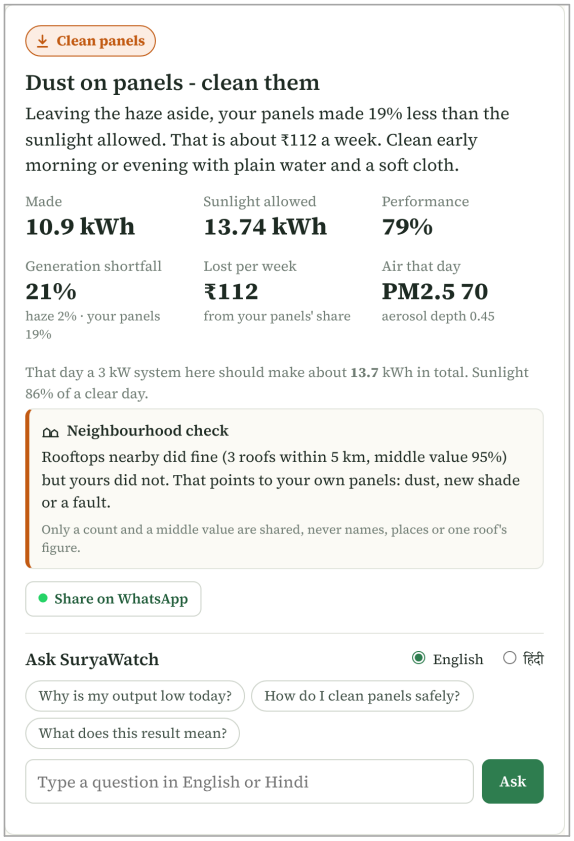
SuryaWatch

Plan rooftop solar. Then make sure it delivers.

The complete guide: what SuryaWatch is, how it works, and how to use every screen.
Environmental Hacks 2026 (WeMakeDevs × AWS) · Track 03: Waste and Energy
Team: Aryan Nirala, Agrim Chaturvedi, Prakhar Srivastava and Anshu Dhaka · Version of 10 October 2026



Watch on a phone



The day's result, with the neighbourhood check

About the pictures in this guide
All screenshots come from a test copy of SuryaWatch filled with made-up sample readings (labelled “demo data”). The satellite map looks blank in them only because the test computer could not load map photos; on your phone or laptop the real satellite view of your roof appears there.

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1. SuryaWatch in one page

SuryaWatch is a free app for Indian homes with rooftop solar, or thinking about it. It works in English and Hindi, on a phone or a laptop, and it needs **no extra hardware**: just a phone camera and the inverter that every solar system already has.

> *"Plan promises you a number of units a year. Watch makes sure you actually get them."*

The problem

- Rooftop solar is spreading fast in India. Under PM Surya Ghar the government has subsidised over 50 lakh home installations, with a target of 75 lakh homes by December 2026. Once installed, almost nobody checks whether the panels produce what they should.
- When output is low, there are three very different reasons: **smog or haze** (nothing to fix), **dust** on the panels (clean them) or a **fault** (call the installer). The inverter only shows a number, so owners guess. They wash panels on smoggy days, wasting water, or miss a real fault for weeks.
- Air pollution and dust can cut solar output in India by 17–25% (a Duke University and IIT Gandhinagar study). In Delhi the October–November smog is the worst time of year for this.
- Before buying, families do not know the right size, the real cost after subsidies, or how fast it pays back.

What SuryaWatch does

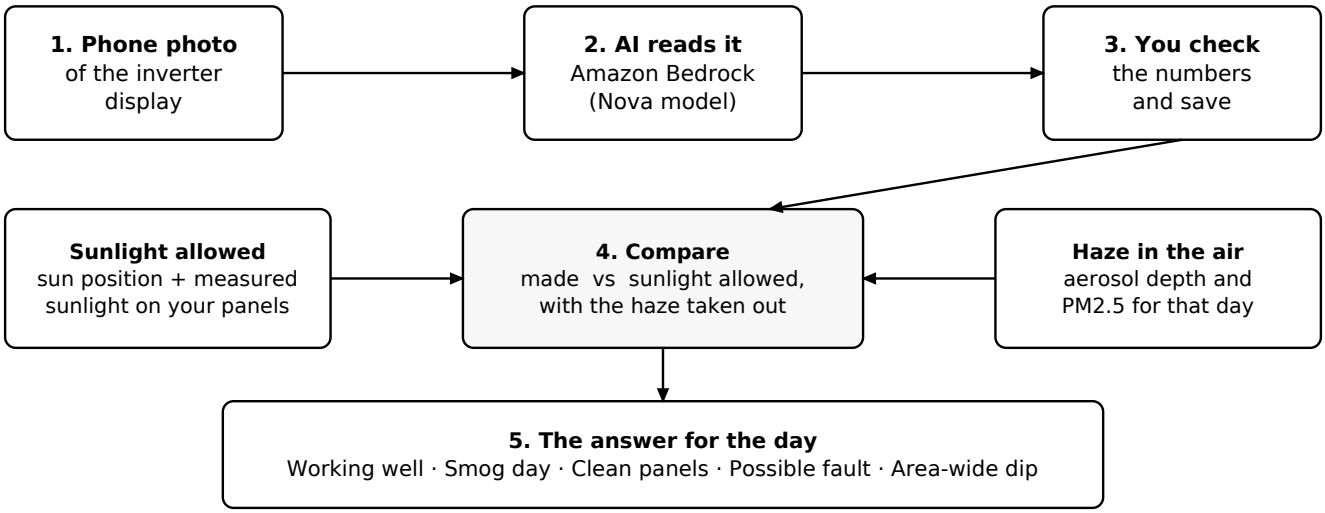
Part	Who it is for	What it gives you
Plan	A home without solar	Draw your roof on the map and enter your electricity use. You get the right system size, what you pay after PM Surya Ghar and Delhi subsidies, the monthly benefit, the payback time and the CO_2 avoided.
Watch	A home with solar	Photograph the inverter display. AI reads it, SuryaWatch compares it with what that day's sunlight allowed, and tells you whether a shortfall is smog, dust or a fault, and what it costs you.
Report	Owner and installer	A printable report for any period: performance, promised vs actual units, fault messages, cleanings, and clear requests for the installer.
Impact	Everyone (public)	Live totals for all rooftops: units made, CO_2 avoided, smog days explained, faults caught, roofs planned.

What makes it different

- **Smog, dust or fault, not just a number.** It uses the day's air-quality data to take the haze out before judging your panels.
- **Neighbourhood check.** If rooftops nearby dipped on the same day, the cause is the sky, not your panels.
- **Best day to clean.** It learns how fast dust builds up on your roof and picks the cheapest day to clean, or tells you to wait for rain.
- **Promised vs got.** It checks the units you actually receive against the installer's promise or the plan.
- **Simple for everyone.** A phone photo, plain-language answers, Hindi at the tap of a button, WhatsApp sharing.

2. How SuryaWatch works

Every check follows the same five steps. You do step 1 and step 3; SuryaWatch does the rest.

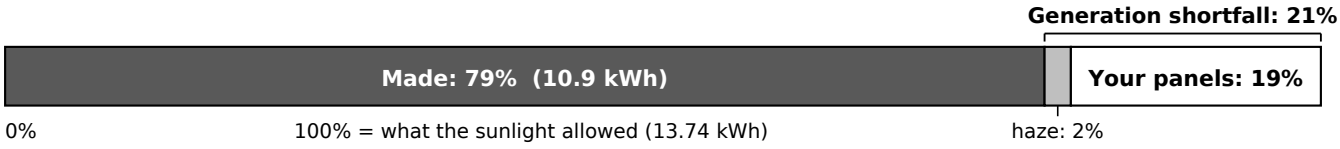


Step by step

1. **Photo.** You photograph the inverter display: the power right now, today's energy (E-Today) and the lifetime total (E-Total). The photo's own clock gives the time it was taken.
2. **AI reads it.** Amazon Bedrock (Amazon's Nova model) reads the numbers and the inverter's status message, even on small two-line screens.
3. **You check.** The numbers appear in boxes. Fix anything that is wrong, then save.
4. **Compare.** SuryaWatch works out how much your system *should* have made: it calculates where the sun was over your roof, takes the sunlight actually measured that day for your panels' tilt and direction (Open-Meteo), and multiplies by your system size. It also reads that day's haze (aerosol optical depth) and PM2.5, and estimates how much sunlight the haze blocked.
5. **Answer.** It compares what you made with what the sunlight allowed, takes the haze out, checks your neighbours, and gives one clear result with what to do.

The numbers, in plain words

Take the dusty day in the screenshot on the cover: the sunlight allowed **13.74 kWh** and the panels made **10.9 kWh**.



Number	Meaning	In the example
Sunlight allowed	What a clean, healthy system of your size should have made with that day's real sunlight (before haze).	13.74 kWh
Performance	Made ÷ sunlight allowed.	10.9 ÷ 13.74 = 79%
Generation shortfall	100% − performance: the part of the possible output you did not get. Performance + shortfall always add up to 100%.	21%
…from haze	The part the smog or haze in the air explains. Not your fault.	2%

Number	Meaning	In the example
…from your panels	The rest: dust, new shade or a fault on your roof. This is what costs you money.	19% → about ₹112 a week

How the result is decided

SuryaWatch looks at your **performance after the haze is taken out** (what you made compared with what the hazy sky allowed):

Situation	Result
The inverter screen shows an error or fault code	**Possible fault**
92% or more of what the hazy sky allowed, and the haze was heavy (it cut 8% or more)	**Smog day** – panels fine
92% or more, clear air	**Working well**
Far below your own recent days (under 60% of them) on a day that was not cloudy	**Possible fault**
Below 92%, and no cleaning or rain for 3 days or more	**Clean panels** (dust)
Below 92% although cleaned or rained on in the last 3 days	**Check system** (shade, loose cable)
Dust or check, but most SuryaWatch roofs within 5 km also dipped that day	**Area-wide dip** – panels fine

A day is judged on its **evening** reading (after 6 PM, or once almost all of the day's sunlight has passed). Earlier readings give a “so far” result.

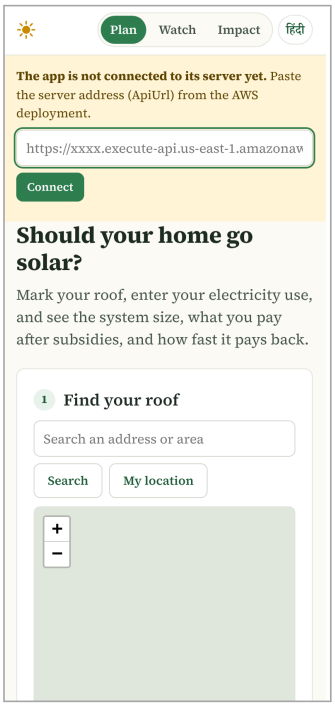
3. Getting started

Opening SuryaWatch

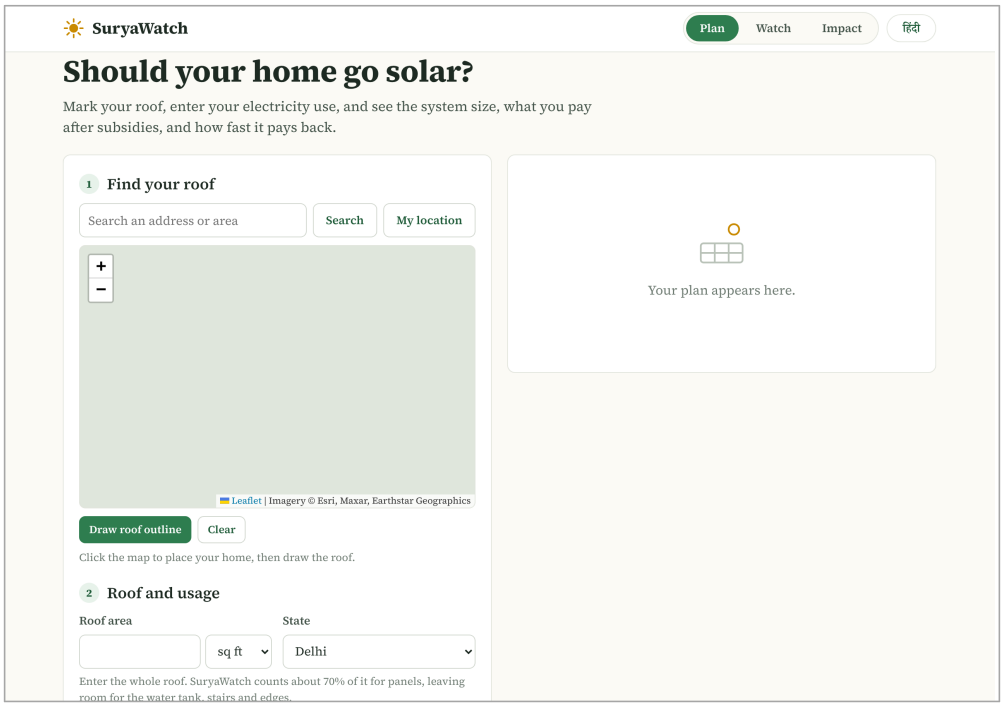
- **Website:** open the SuryaWatch web address (the WebUrl from the AWS deployment) in any browser.
- **Android app:** install the SuryaWatch APK. The first time it opens, it asks for the **server address** (the ApiUrl from the AWS deployment). Paste it and tap **Connect**. The app remembers it; you can change it later from the “Server: … · Change” line at the bottom of the app.
- **Laptop test copy:** run `python scripts/local_server.py --mock-ai` in the project folder and open http://localhost:8080 (see section 11).

The top bar has three tabs and a language button:

- **Plan**: should your home go solar?
- **Watch**: is your solar working?
- **Impact**: what SuryaWatch has done so far.
- **The Hindi button** हिंदी (it reads “English” while in Hindi): switches the whole app, including results, messages and WhatsApp text. Your choice is remembered.



First launch of the Android app



The app on a laptop: the Plan tab

4. Plan: should your home go solar?

Plan takes about two minutes. You need your roof's location and one electricity bill (or your monthly units).

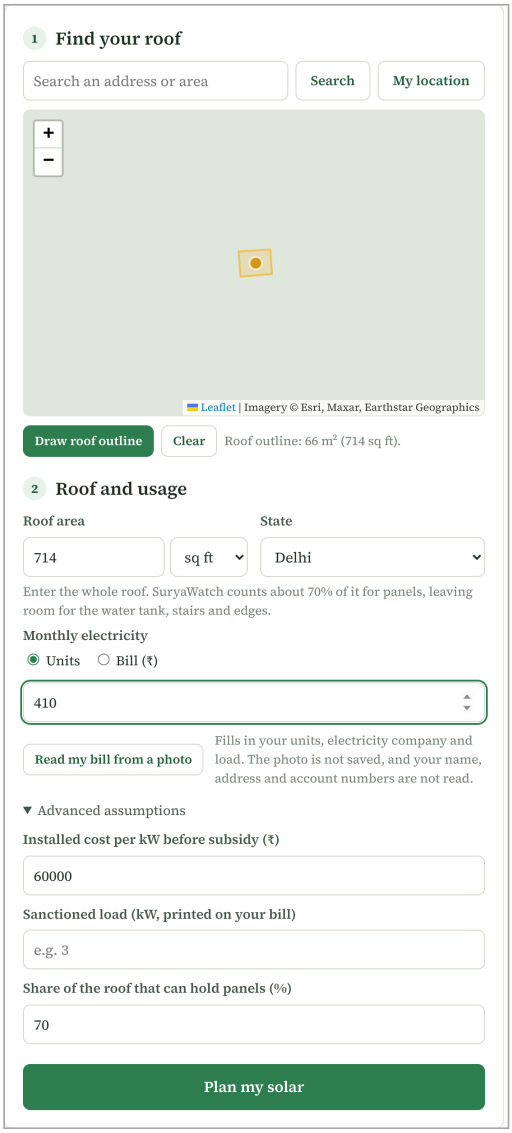
Step 1 · Find your roof

- Type your address or area and tap **Search**, or tap **My location**, or click your home on the map.
- Tap **Draw roof outline** and tap each corner of your roof. The map zooms in for you. Tap **Draw** again (or double-tap) to finish. The roof area fills in by itself.
- **Clear** removes the outline if you want to redraw it.

Step 2 · Roof and usage

- **Roof area** in sq ft or m². Enter the whole roof; SuryaWatch counts about 70% of it for panels.
- **State**: Delhi gets the extra Delhi subsidy and incentive.
- **Monthly electricity**: units from your bill, or the bill amount in ₹.
- **Read my bill from a photo**: AI reads the units, your electricity company and sanctioned load. The photo is not saved, and your name, address and account numbers are not read.
- **Advanced assumptions** (optional): installed cost per kW, sanctioned load, and the share of the roof that can hold panels.

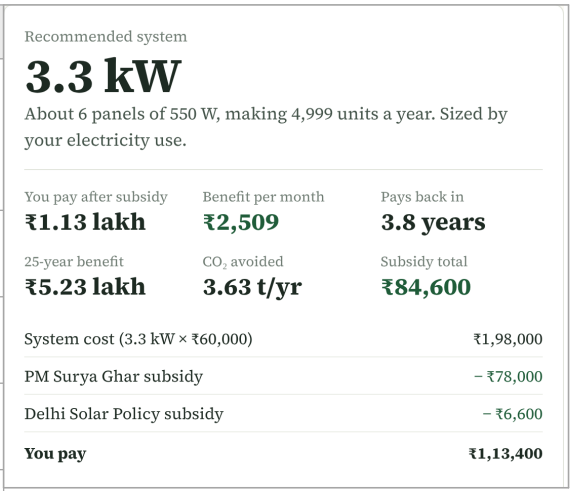
Step 3 · Tap **Plan my solar**.



The Plan form with a roof outline

Reading your plan

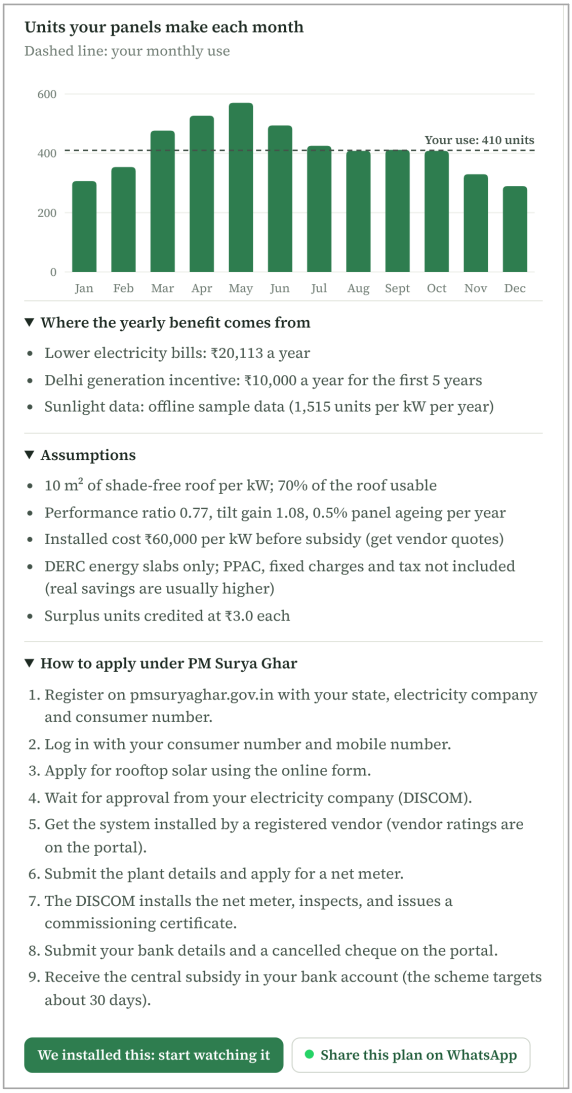
Number	What it means
Recommended system	Size in kW, built from whole 550 W panels (6 panels = 3.3 kW). Sized by your use or by your roof, whichever is smaller.
You pay after subsidy	Installed cost minus the PM Surya Ghar and Delhi subsidies.
Benefit per month	Average bill saving plus the Delhi generation incentive.
Pays back in	Years until the benefit covers what you paid.
25-year benefit	Total benefit over the panels' life, with 0.5% ageing a year.
CO_2 avoided	Tonnes a year, at 0.727 kg CO_2 per unit (India's grid average, CEA).
Subsidy total	Central plus Delhi subsidy.



The top of a plan

Below the numbers:

- **Units your panels make each month**: bars for each month, with a dashed line for your use. Summer months make more.
- **Where the yearly benefit comes from**: bill saving and the Delhi generation incentive.
- **Assumptions**: every number SuryaWatch assumed, so nothing is hidden.
- **How to apply under PM Surya Ghar**: the nine official steps, from registering on pmsuryaghar.gov.in to the subsidy reaching your bank.
- **Share this plan on WhatsApp**: sends the summary and a link.
- **We installed this: start watching it**: turns the plan into a Watch rooftop, and the plan's yearly units become the promise SuryaWatch checks.



Monthly chart, assumptions and how to apply

The subsidy rules SuryaWatch uses

Scheme	Rule
PM Surya Ghar (central)	₹30,000 per kW for the first 2 kW, ₹18,000 for the 3rd kW; at most ₹78,000.
Delhi Solar Policy 2024	Extra ₹2,000 per kW, at most ₹10,000. Generation incentive: ₹3 per unit (systems up to 3 kW) or ₹2 per unit (3–10 kW) for 5 years.
Delhi bills (DERC)	Slabs: 0–200 units ₹3, 201–400 ₹4.50, 401–800 ₹6.50, 801–1,200 ₹7, above 1,200 ₹8. Up to 200 units a month is free; 201–400 gets 50% off (up to ₹800).
Net metering	Extra units you send to the grid are credited (SuryaWatch assumes ₹3 each).

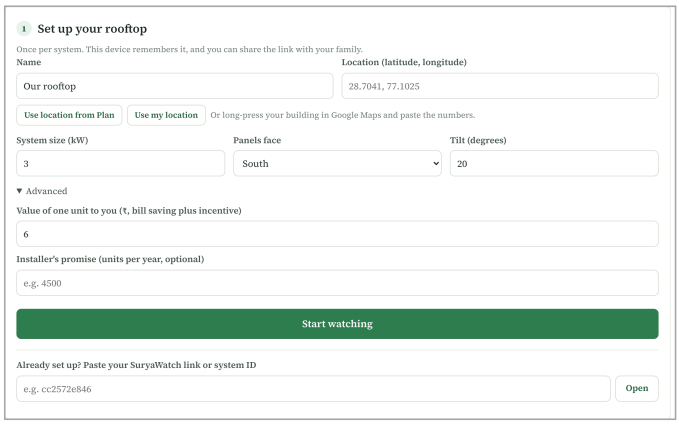
These are estimates to compare options. Always get quotes from registered vendors before buying.

5. Watch: is your solar working?

5.1 Set up your rooftop (once)

Fill this in once per solar system:

- **Name** of the rooftop.
- **Location**: latitude and longitude. Tap **Use location from Plan** or **Use my location**, or long-press your building in Google Maps and paste the numbers.
- **System size** in kW (on the inverter label or the installer's bill), which way the panels **face**, and their **tilt**.
- **Advanced**: what one unit is worth to you, and the installer's promised units a year.
- Tap **Start watching**. This device remembers the rooftop. To open it on another phone, use **Copy link** or paste the system ID under “Already set up?”.



First-time setup

5.2 The dashboard



Top of the dashboard: rooftop name, day picker and buttons

The arrows and the date box move between days. **Report for installer** opens the report (section 6), **Copy link** copies a link to this rooftop for your family, and **Other system** switches rooftops. Below come, in order: the day's result, Add readings, Next two days, Dust and the best day to clean, Promised vs got, the charts, the day-by-day calendar, Email alerts, and the day's readings.

5.3 Adding readings

Three ways to add a reading:

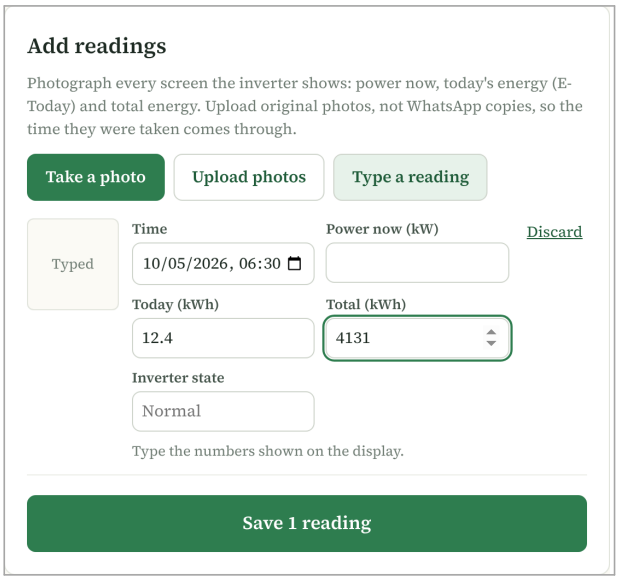
- **Take a photo**: opens the phone camera.
- **Upload photos**: pick one or many photos from the gallery.
- **Type a reading**: enter the numbers yourself.

What to photograph

- Every screen the inverter cycles through: **power now** (kW), **today's energy** (E-Today, kWh) and **total energy** (E-Total, kWh), plus any status or error message.
- Best times: around **1 PM** (strongest sun) and **after 5:45 PM** (the day's total).
- Use the original photos, not WhatsApp copies, so the time the photo was taken comes through.

Then

- Each photo appears as a row. AI fills in the boxes; check them and correct anything wrong.
- Tap **Save readings**. The day's result updates straight away.

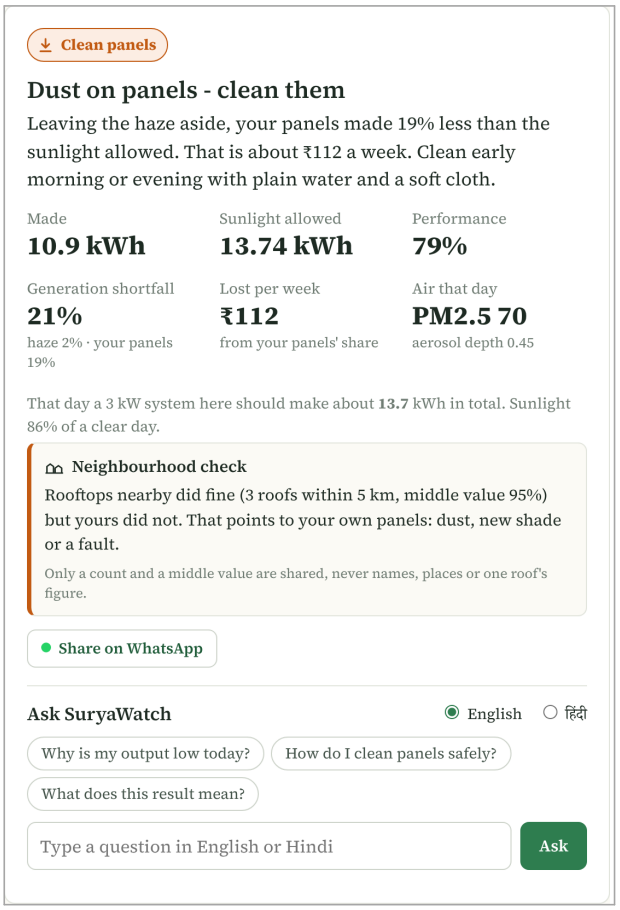


A reading waiting to be saved

5.4 The day's result

The most important card. From the top:

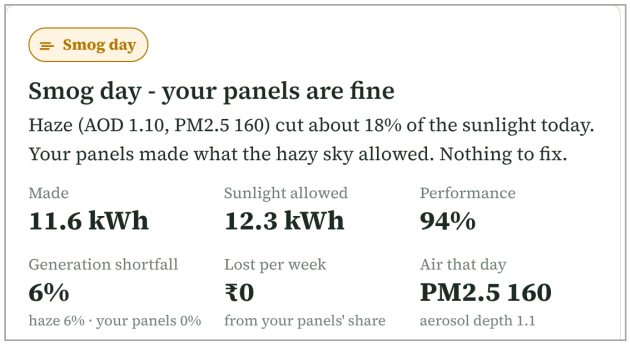
- **Badge and title**: the result in a few words (see 5.5).
- **Message**: why, and what to do.
- **Made** and **Sunlight allowed** in kWh.
- **Performance** and **Generation shortfall**, which always add up to 100%. Under the shortfall: how much was haze and how much was your panels.
- **Lost per week**: what your panels' share costs you in rupees if it continues.
- **Air that day**: PM2.5 and aerosol depth (haze).
- **Neighbourhood check** (5.6), **Share on WhatsApp**, and **Ask SuryaWatch** (5.14).



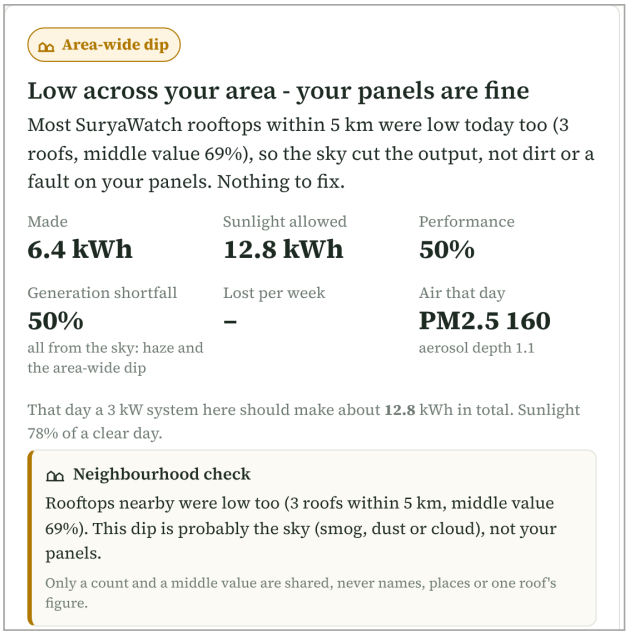
A dusty day

5.5 Every result you can see

Result	What it means	What to do
Working well	The panels made what the sunlight allowed.	Nothing.
Smog day	Output was low because haze blocked sunlight. The panels made what the hazy sky allowed.	Nothing. Do not wash the panels for this.
Clean panels	After taking the haze out, output is still low and there has been no cleaning or rain for 3+ days: dust. The message gives the rupees lost a week.	Clean early morning or evening with plain water and a soft cloth, or wait for forecast rain.
Possible fault	The inverter showed an error, or output suddenly fell far below your recent days.	Note the code. Switch the inverter off and on once if your installer allows it. Call the installer.
Check system	Low although the panels were cleaned or rained on recently.	Look for new shade, a loose cable or inverter warnings.
Area-wide dip	Most SuryaWatch rooftops within 5 km were low on the same day.	Nothing: the sky (smog, dust storm or cloud) caused it. No cleaning alert is sent.
No reading yet / Too early	No photo for this day yet, or it is before about 10 AM.	Add a reading later in the day.



A smog day: haze explains the shortfall



An area-wide dip: every roof nearby was low

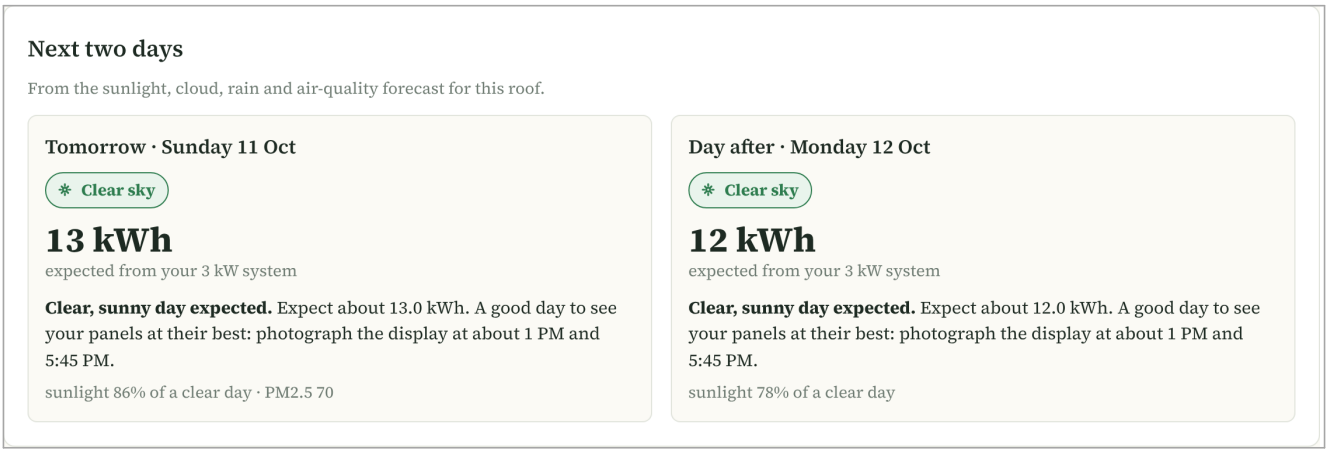
5.6 Neighbourhood check

Under every result, SuryaWatch compares your day with other SuryaWatch rooftops within 5 km:

- **They were low too**: the dip is probably the sky (smog, dust in the air or cloud), not your panels. A dust or check result becomes **Area-wide dip**, and no cleaning alert is sent.
- **They did fine but yours did not**: the problem is on your own roof (dust, new shade or a fault).
- **About the same**: your roof did as well as its neighbours.
- **No neighbours yet**: an **Invite neighbours on WhatsApp** button appears.

Privacy: only a count and a middle value from at least two neighbours are ever shown. Never a name, a place or one roof's figure. Sample (demo) rooftops are compared only with other sample rooftops.

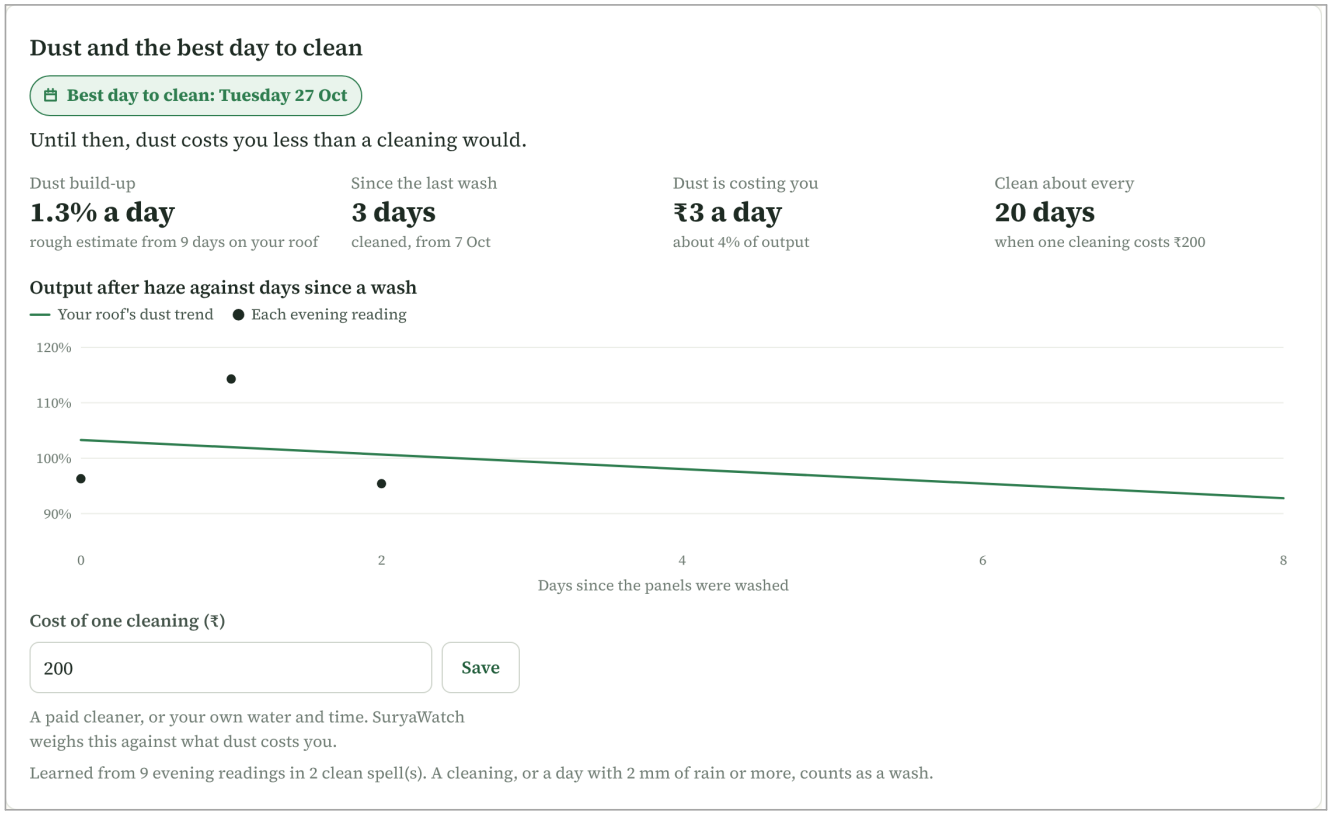
5.7 Next two days



The forecast for tomorrow and the day after

For tomorrow and the day after: the units your system will likely make, and one of **Clear sky**, **Hazy**, **Heavy smog**, **Cloudy** or **Rain**, with advice. Rain of 5 mm or more is a free wash, so you can skip cleaning before it. When heavy smog is coming, SuryaWatch warns you that low output will not mean dirty panels.

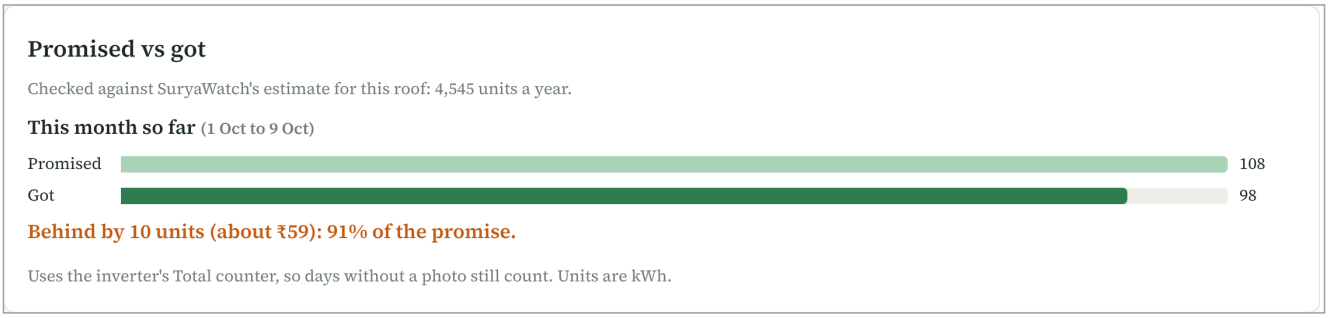
5.8 Dust and the best day to clean



The dust card

- **Dust build-up**: how much output your roof loses per day of dust. SuryaWatch learns this from your own evening readings after each wash (a cleaning, or a day with 2 mm of rain or more). Until it has enough days (about 5 evening readings over 4+ days after a wash) it uses a typical 0.4% a day, measured on an IIT Bombay rooftop in the dry season.
- **Since the last wash**: days since the last cleaning or rain.
- **Dust is costing you**: rupees a day right now.
- **Clean about every**: the gap between cleanings that keeps the total cost lowest. It weighs what dust costs you, which grows every day, against the **Cost of one cleaning** you enter (₹200 by default: a paid cleaner, or your own water and time). It is never more than a month.
- **Best day to clean**: the date that gives. If at least 5 mm of rain is forecast around then, it says **Wait for the rain** instead, which saves water and money.
- The chart shows each evening reading against days since a wash, with your roof's dust trend line.

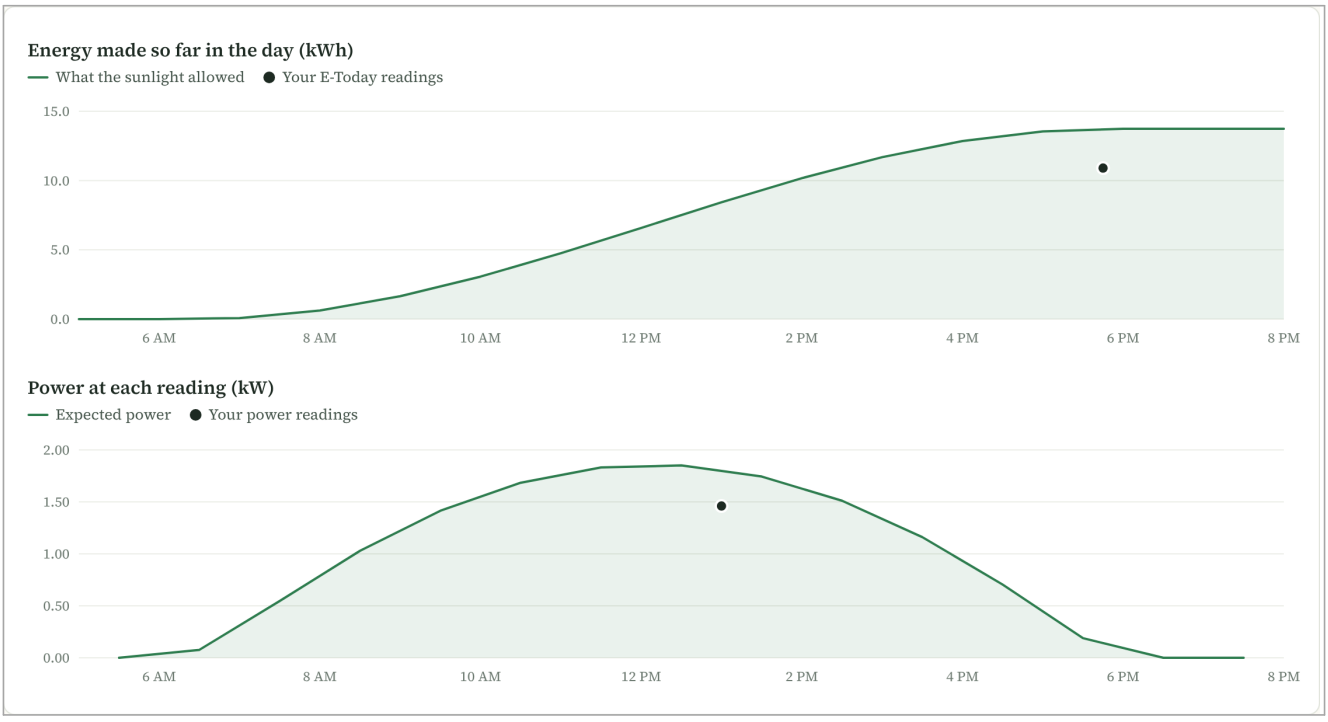
5.9 Promised vs got



Promised units against the units actually made

SuryaWatch checks the units you really received against a promise: the installer's yearly figure if you entered one, your SuryaWatch plan, or SuryaWatch's own estimate for your roof. It uses the inverter's lifetime counter (E-Total), so days without a photo still count. You see this month so far and the whole period since tracking began, with the gap in units and rupees. To start it, add two photos of the Total (E-Total) screen taken a day or more apart.

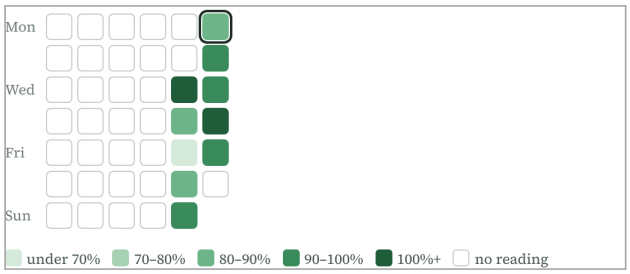
5.10 Charts for the day



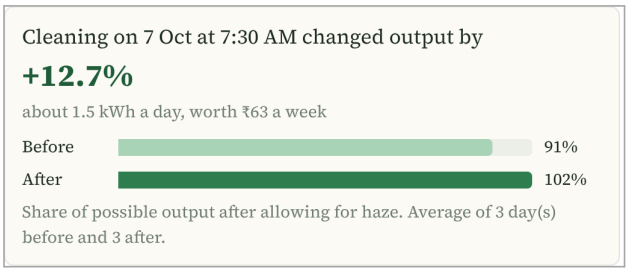
Energy through the day and power at each reading

The shaded curve is what the sunlight allowed through the day; the dots are your readings. Hover or tap for exact values. Dots below the curve mean a shortfall at that time of day.

5.11 Day-by-day calendar and cleaning effect



Each square is one day

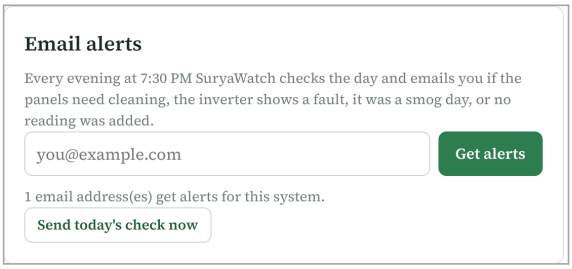


What the last cleaning changed

- **Calendar**: one square per day for the last five weeks. Darker green means your panels gave more of the possible output (after haze). Hollow squares had no reading. A red ring marks a possible fault. Tap a square to open that day.
- **Cleaning effect**: after you press **We cleaned the panels today**, SuryaWatch compares the 3 days before with the 3 days after, and shows the gain in %, kWh a day and rupees a week. This is the proof that cleaning was worth it.

5.12 Email alerts

- Enter an email and tap **Get alerts**. On AWS you will get one confirmation email first; click its link.
- Every evening at **7:30 PM** SuryaWatch checks the day and emails you if the panels need cleaning (with the best day to clean), the inverter showed a fault, it was a smog day, or no reading was added.
- It also warns you the evening before heavy smog or rain.
- The email includes the neighbourhood comparison and a line in Hindi.
- **Send today's check now** runs the check straight away.



Email alerts

5.13 Readings on this day



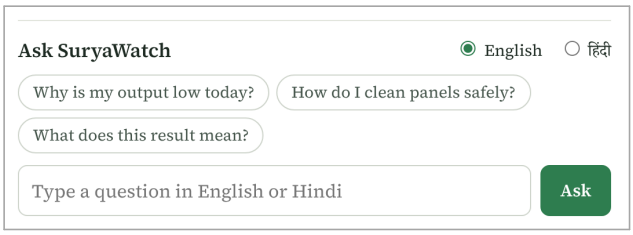
The day's readings

Every reading saved for the day, with where it came from (photo or typed). **Remove** deletes a wrong one. **We cleaned the panels today** records a cleaning, which the dust card and the cleaning effect use.

5.14 Ask SuryaWatch

A helper that answers questions about your plan or your day, in English or Hindi. Tap a suggested question or type your own, for example “Why is my output low today?”. Choose English or Hindi for the answer.

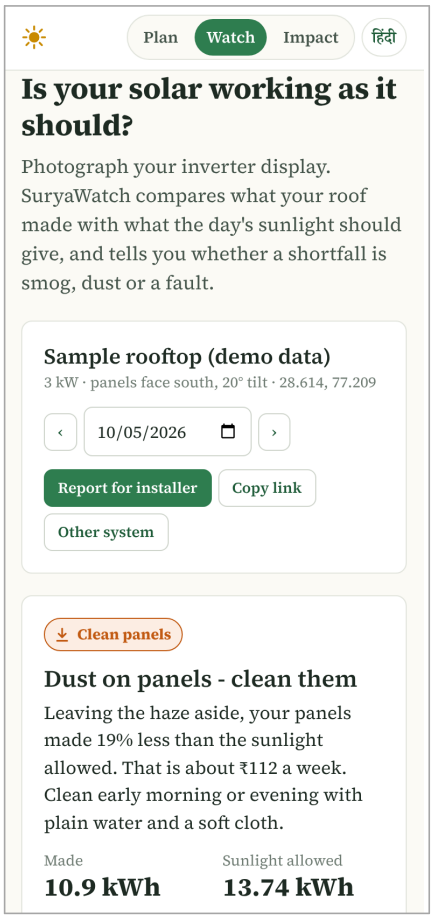
It is powered by Amazon Bedrock and uses your own numbers plus checked facts about PM Surya Ghar and the Delhi Solar Policy. It works once SuryaWatch is running on AWS.



Ask SuryaWatch

5.15 On a phone

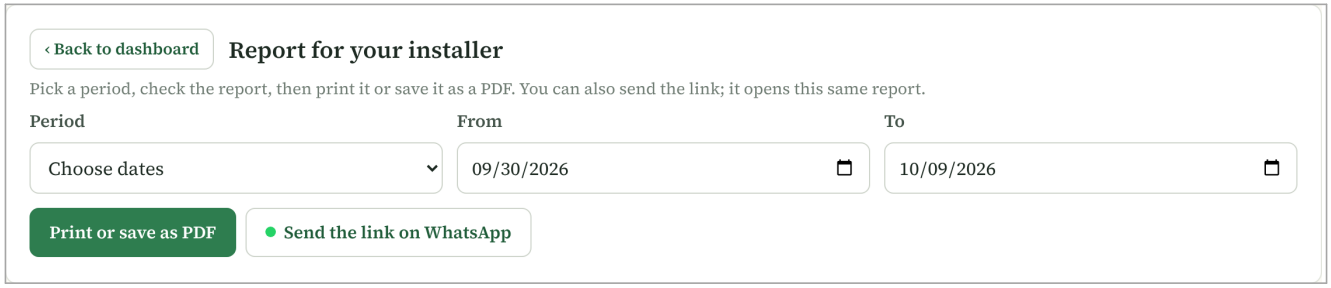
Everything works the same on a phone. Cards stack one below another, and **Take a photo** opens the camera directly. On narrow screens the top bar shows only the sun icon, the three tabs and the language button.



Watch on a phone

6. Report for your installer

Tap **Report for installer** on the dashboard.

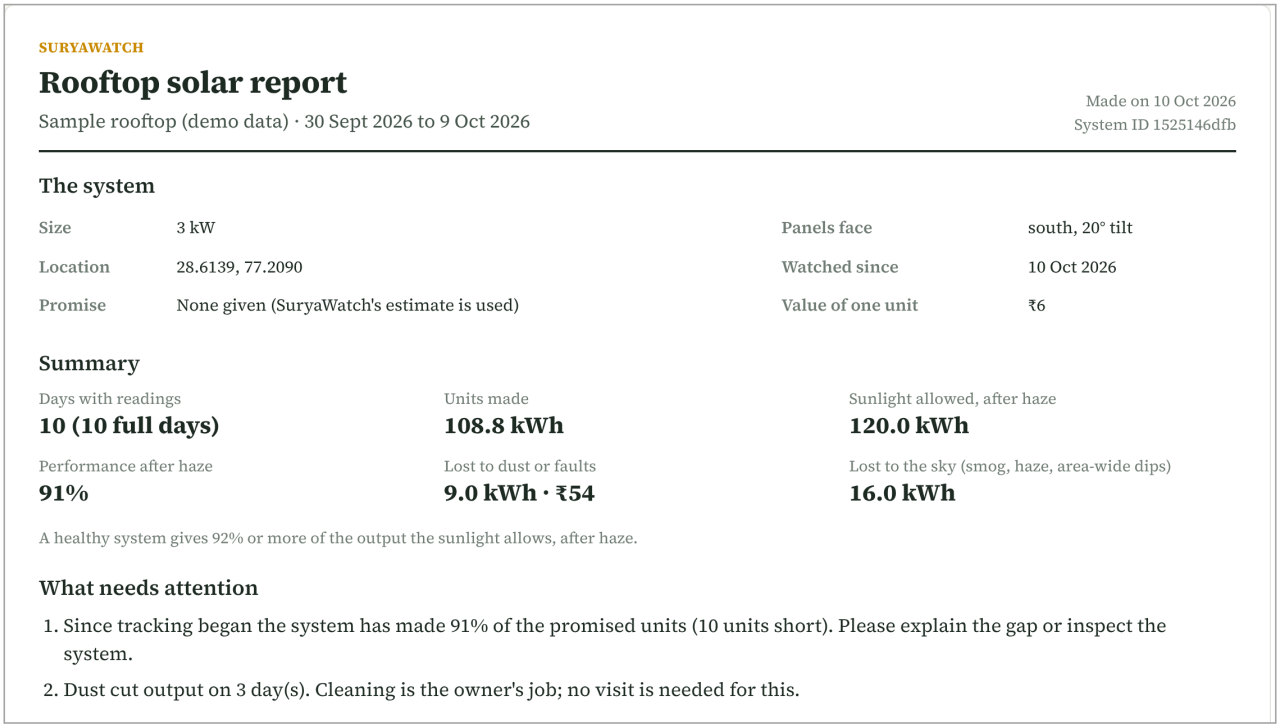


Choosing the report period

- Pick a period: last 30 days, last 7 days, this month, last month, or your own dates (up to 92 days).
- **Print or save as PDF**: prints two black-and-white A4 pages. In the Android app this opens the same report in your browser to print or save it there.
- **Send the link on WhatsApp**: sends a summary and a link that opens this same report.

What the report contains

- The system's details, and a summary: units made, sunlight allowed after haze, performance, units lost to dust or faults (kWh and ₹), and units lost to the sky (smog, haze, area-wide dips).
- **What needs attention**: clear requests for the installer, such as inspecting a fault, explaining a gap with the promised units, or checking wiring. Dust is noted as the owner's job.
- Promised vs got, and the inverter's Total counter at the start and end of the period.
- Every fault message seen on the photos, with date and time; the cleanings and their effect.
- A day-by-day chart and table, and how the report was made.



The report on screen

SURYAWATCH

Rooftop solar report

Sample rooftop (demo data) · 30 Sept 2026 to 9 Oct 2026

Made on 10 Oct 2026
System ID 1525146dfb

The system

Size	3 kW	**Panels face**	south, 20° tilt
Location	28.6139, 77.2090	**Watched since**	10 Oct 2026
Promise	None given (SuryaWatch's estimate is used)	**Value of one unit**	₹6

Summary

Days with readings
10 (10 full days)

Units made
108.8 kWh

Sunlight allowed, after haze
120.0 kWh

Performance after haze
91%

Lost to dust or faults
9.0 kWh · ₹54

Lost to the sky (smog, haze, area-wide dips)
16.0 kWh

A healthy system gives 92% or more of the output the sunlight allows, after haze.

What needs attention

1. Since tracking began the system has made 91% of the promised units (10 units short). Please explain the gap or inspect the system.
2. Dust cut output on 3 day(s). Cleaning is the owner's job; no visit is needed for this.

Promised vs got

Period	Promised	Got	Share
This month so far (1 Oct to 9 Oct)	108	98	91%
Since tracking began (1 Oct to 9 Oct)	108	98	91%

Units are kWh.

Inverter Total counter

4,011 kWh on 30 Sept, 5:45 PM to 4,109 kWh on 9 Oct, 5:45 PM: 98 kWh in between.

Inverter faults and messages

No fault messages on the photos in this period.

Cleanings

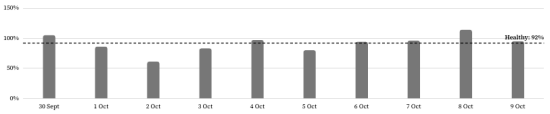
7 Oct 2026, 7:30 AM

Cleaning on 7 Oct 2026 changed output from 91% to 102% of possible (+12.7%).

Printed report, page 1

Day by day

Share of possible output each full day, after haze (%)



Date	Made (kWh)	Sunlight allowed (kWh)	Haze	Performance after haze	Result
Wed, 30 Sept	11.2	13.0	18%	105%	Smog day
Thu, 1 Oct	12.0	14.1	1%	86%	Clean panels
Fri, 2 Oct	6.4	12.8	18%	61%	Area-wide dip
Sat, 3 Oct	11.4	13.9	1%	83%	Clean panels
Sun, 4 Oct	10.1	12.6	18%	97%	Smog day
Mon, 5 Oct	10.9	13.7	1%	80%	Clean panels
Tue, 6 Oct	9.7	12.5	18%	94%	Smog day
Wed, 7 Oct	12.9	13.6	1%	96%	Working well
Thu, 8 Oct	11.6	12.3	18%	114%	Smog day
Fri, 9 Oct	12.6	13.4	1%	95%	Working well

How this report was made

- Readings come from photos of the inverter display, read by AI and checked by the owner, or typed in.
- Expected output uses the sun's position over this roof, measured sunlight from Open-Meteo, and the system's size, tilt and direction.
- Haze comes from the air-quality data (aerosol optical depth). Output lost to haze is not counted against the panels.
- Each day is judged on its evening E-Today reading; a part day had no evening reading.
- These are estimates to guide a check-up. A site visit is the final word.

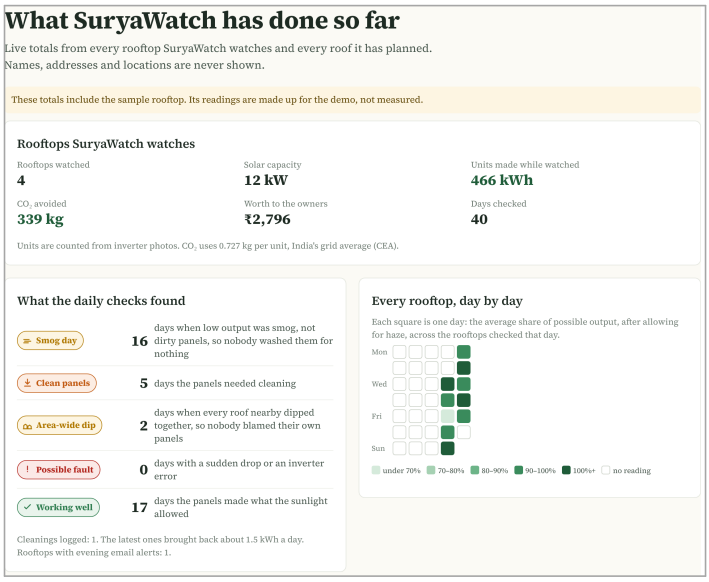
Made with SuryaWatch. Open this report online: http://localhost:8097/?system=1525146dfb&from=2026-09-30&to=2026-10-09#report

Printed report, page 2

7. Impact: what SuryaWatch has done

The **Impact** tab is public. It shows live totals across every rooftop:

- Rooftops watched, their kW, units made while watched, CO_2 avoided, their worth to the owners, days checked.
- What the daily checks found: smog days explained (no needless washing), dust warnings, area-wide dips, possible faults, healthy days, and cleanings with the output they brought back.
- A green calendar for all rooftops, and a bar chart of units made each day.
- Roofs planned: how many, the kW they could hold, units a year, CO_2 a year and subsidy.
- No names, IDs, addresses or locations are ever shown. Each roof planned counts once.
- Sample (demo) rooftops are left out unless you tick the box; then a yellow note says the totals include made-up data.



Impact, with the sample rooftops included



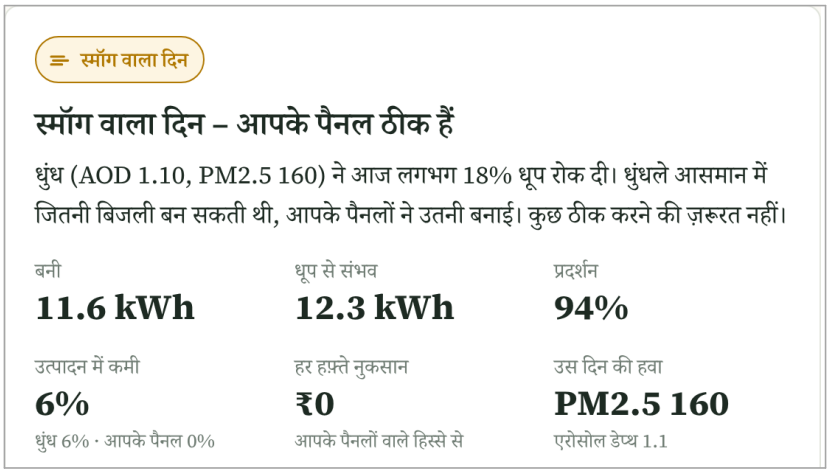
Units made each day, all rooftops

8. SuryaWatch in Hindi

Tap the Hindi button हिंदी in the top bar. Everything switches: buttons, results, messages, the forecast, the plan's notes, the report, the Impact page and the WhatsApp text. Tap **English** to switch back. Numbers, kW, kWh and PM2.5 stay the same. The choice is remembered on that phone or laptop.



The top of the Watch tab in Hindi



A smog day result in Hindi

9. A simple routine for owners

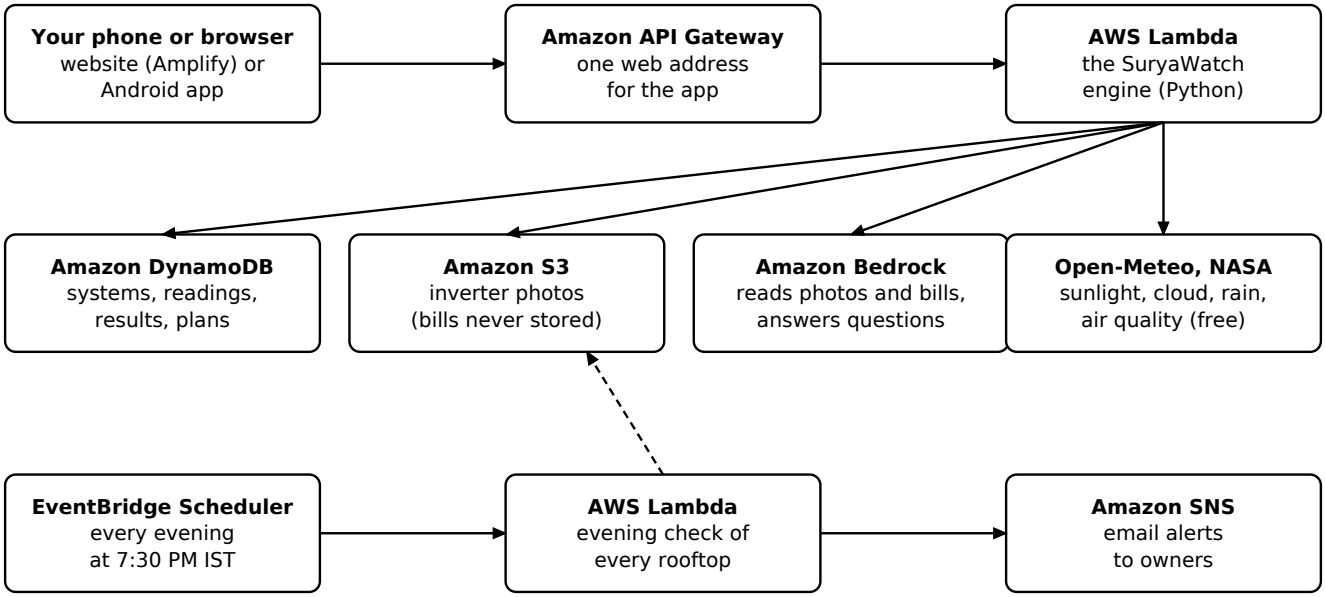
When	What to do	Time
Every day (or most days)	Photograph the inverter after 5:45 PM (E-Today and E-Total). Add a 1 PM photo of the power screen if you can.	1 minute
When the evening email or the result says **Clean panels**	Check the dust card: clean on the best day, or wait if rain is forecast. Then press **We cleaned the panels today**.	Cleaning time
When it says **Smog day** or **Area-wide dip**	Nothing. Do not wash the panels.	–
When it says **Possible fault**	Photograph the error screen, then send the installer report.	5 minutes
Once a month	Look at **Promised vs got**. If it stays behind, send the installer report.	2 minutes

Why the evening photo matters most

The evening E-Today number is the day's full total, so SuryaWatch can give a final result for the day. The E-Total number lets it check promised units even on days you miss.

10. Behind the scenes: AWS and data

SuryaWatch runs on Amazon Web Services. Nothing runs on a server you have to look after.



AWS service	What it does in SuryaWatch
AWS Amplify Hosting	Serves the website.
Amazon API Gateway	One web address the app talks to.
AWS Lambda	The SuryaWatch engine in Python: planning, expected output, results, dust, reports, impact.
Amazon DynamoDB	Stores rooftops, readings, daily results, events and plans.
Amazon S3	Stores inverter photos (privately). Bill photos are never stored.
Amazon Bedrock (Nova)	Reads inverter displays and bills; answers questions in English and Hindi.
Amazon EventBridge Scheduler	Starts the evening check at 7:30 PM India time.
Amazon SNS	Sends the email alerts.
AWS SAM	Builds and deploys all of the above with one command.

Free public data

Source	Used for
Open-Meteo forecast	Measured and forecast sunlight on your panels' tilt, cloud, temperature and rain.
Open-Meteo air quality	PM2.5, AQI and aerosol optical depth (haze) for every hour.
NASA POWER	Long-term monthly sunlight, used by the planner.
CEA	India's grid emission factor: 0.727 kg CO_2 per unit.

Privacy

- Bill photos are read in memory and never saved; names, addresses and account numbers are not read.

- A rooftop is reached only through its own link or ID.
- The Impact page and the neighbourhood check show only totals, counts and middle values.
- Sample data is always labelled “demo data” and kept out of public totals.

11. Running SuryaWatch

A. On a laptop, for testing (no AWS needed)

Install Python 3, then in the suryawatch folder:

```
python scripts/local_server.py --mock-ai
```

Open http://localhost:8080. Photo and bill reading give test values, and the helper gives a test answer, because the real AI runs on AWS. To fill the app with labelled sample data (a sample rooftop with three sample neighbours), run this in a second window:

```
python scripts/sample_data.py
```

Add `--offline` to the first command to work without internet (made-up weather).

B. On AWS (Agrim)

Agrim follows the AWS guide: in AWS CloudShell, clone the repository and run

```
bash scripts/deploy.sh
```

At the end it prints the **ApiUrl** (server address) and the **WebUrl** (website).

C. The Android app (APK)

Build the single-file app and turn it into an APK with any HTML-to-APK converter:

```
python scripts/build_single_html.py
```

This writes `dist/SuryaWatch.html`. Without addresses, the app asks for the server address on first launch. To build the addresses in instead:

```
python scripts/build_single_html.py --api <ApiUrl> --web <WebUrl>
```

In the converter, allow internet, file upload and camera, location, and opening links in the browser (for WhatsApp sharing and printing).

12. Words used in SuryaWatch

Word	Meaning
kW (kilowatt)	Size of a solar system, or power at one moment. A 3 kW system makes up to about 3 kW at noon.
kWh or unit	Energy. One unit on your electricity bill = 1 kWh.
E-Today	The inverter's count of energy made today (kWh).
E-Total	The inverter's lifetime count of energy made (kWh). It only ever goes up.
Inverter	The box that turns the panels' power into household electricity and shows the numbers.
Sunlight allowed	What a healthy system of your size should have made with that day's real sunlight.
Performance	Made ÷ sunlight allowed.
Generation shortfall	100% − performance: the part of the possible output you did not get.
Haze, AOD	Fine particles in the air that block sunlight. Aerosol optical depth (AOD) measures it; Delhi's smog often pushes it above 1.
PM2.5	Fine-particle pollution in the air (µg/m³).
Performance ratio	How much of the sunlight's energy a real system delivers after its own losses (SuryaWatch assumes 0.85 for daily checks, 0.77 for planning).
DISCOM	Your electricity company (in Delhi: BRPL, BYPL or TPDDL).
Net metering	A two-way meter: extra solar units go to the grid and are credited on your bill.
Sanctioned load	The maximum load on your connection, printed on your bill. Some DISCOMs limit solar to it.
Generation incentive (GBI)	Delhi pays ₹2–3 for every solar unit you make, for 5 years.
PM Surya Ghar	The central government scheme that subsidises home rooftop solar (pmsuryaghar.gov.in).

13. Questions and problems

“The app is not connected to its server yet.”
The AWS part is not running yet, or the address is wrong. Ask Agrim for the ApiUrl and paste it in the yellow box, or tap Change at the bottom of the app. On a laptop, start local_server.py and use http://localhost:8080.

The map is blank or grey.
Map photos need internet. Check the connection; you can still type the roof area.

The time on my reading is wrong.
WhatsApp copies lose the time a photo was taken. Use the original photo, or correct the time box before saving.

AI read a number wrongly.
Correct it in the box before you save. Hold the phone straight and avoid glare on the display.

It says Smog day, but my output is low.
That is the point: the haze in the air blocked sunlight, so your panels are fine. Washing them will not help that day.

Why does the plan show 3.3 kW and not 3 kW?
Plans are sized in whole 550 W panels: 6 panels × 550 W = 3.3 kW.

Why “Clean about every 20 days” when my neighbour cleans weekly?
It depends on your roof's dust rate, your system size, what a unit is worth to you and what a cleaning costs you. Change the cleaning cost on the dust card to see the effect.

Print does nothing in the Android app.
Phone apps cannot print. The button opens the report in your browser; print or save it as PDF from there.

Do neighbours see my numbers?
No. They see only a count and a middle value from at least two roofs, never your name, place or your own figure.

Sources for the figures in this guide
PM Surya Ghar progress and target: Outlook Business and SolarQuarter (2026). Air pollution and dust cutting Indian solar output by 17–25%: Duke University and IIT Gandhinagar study (2017). Typical dust loss of 0.4% a day: IIT Bombay rooftop, pv magazine India (July 2021). Subsidies: PM Surya Ghar (PIB, March 2025) and the Delhi Solar Policy 2024. Tariffs: DERC. Grid emission factor: CEA CO_2 baseline database (FY 2023-24).